Neha G. Pusalkar

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Research Statement

My research develops scalable, computationally efficient coalition formation algorithms for heterogeneous multi-robot systems and large-scale collectives exceeding 1000 robots. I employ game-theoretic and sequential decision-making methods to design distributed coordination frameworks for dynamic, uncertain environments. Additionally, I investigate preemptive coalition reformation mechanisms that enable adaptive coalition restructuring for complex, real-world problems.

Research Interests: **Multi-Robot Task Allocation, Collective Robotics, Game Theory, Distributed A.I., Multi-Objective Optimization, Multi-Robot Planning and Reinforcement Learning.**

Education

Oregon State University, Corvallis, OR, USA 2026 (Expected)
Doctor of Philosophy, Robotics and Artificial Intelligence (3.85/4.0)
Dissertation: Scalable Coalition (Re)formation for Multiple Robot Systems and Extremely Large Collectives.
Advisor: Prof. Julie A. Adams (Human Machine Teaming Lab)

University of Michigan, Ann Arbor, MI, USA 2021
Master of Science, Robotics (GPA: 3.60/4.0)
Advisor: Prof. Chad Jenkins (The Laboratory For Progress)
Project PI: Prof. Ram Vasudevan UM Ford Center for Autonomous Vehicles (FCAV)

Visvesvaraya National Institute of Technology, Nagpur, India 2018
Bachelor of Technology, Electronics and Communication Engineering (GPA: 8.78/10.0)

Publications

[P1] **Neha G. Pusalkar**, Julie A. Adams, *Dynamic Multiple Robot Task Allocation: A Literature Review*, IEEE Access, 2026 (*In Review*)

[P2] **Neha G. Pusalkar**, Julie A. Adams, *Preemptive Multiple Robot Coalition Reformation*, International Symposium on Distributed Autonomous Robotic Systems (DARS), 2026

[P3] **Neha G. Pusalkar**, Julie A. Adams, *Extremely Large Collective Coalition Formation: Scalability*, International Conference on Autonomous Agents and Multi-agent Systems (AAMAS), 2026

[P4] **Neha G. Pusalkar**, Julie A. Adams, *Leader-based Coalition Formation for Extremely Large Scale Collectives*, 2025 IEEE Conference on Artificial Intelligence, pp. 701-706, May 2025 (Best Paper Award in Robotics/UAV).

[P5] **Neha G. Pusalkar**, Mark-Robin Giolando, Julie A. Adams, *Decision Support System for Autonomous Underwater Grasping*, Late Breaking Report, Companion of 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI), pp. 198-202, March 2023

[P6] **Neha G. Pusalkar**, Mark-Robin Giolando, Julie A. Adams, *Autonomous Underwater Robot Grasping Decision Support System*, Videos and Demos, Companion of 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI), pp. 887-888, March 2023

[P7] Alphonsus Adu-Bredu, Zhen Zeng, **Neha Pusalkar**, Odest Chadwicke Jenkins, *Elephants Don't Pack Groceries: Robot Task Planning for Low Entropy Belief States*, IEEE Robotics and Automation Letters (RA-L), vol. 7, no. 1, pp. 25-32, Jan. 2022

[P8] Xiaoxiao Du, **Neha Pusalkar**, Ram Vasudevan, Matthew Johnson-Robertson, *Angle-Regulated Transformer Network for Pedestrian Trajectory Prediction*, AI for Autonomous Driving (AI4AD) Workshop,

International Joint Conference on Artificial Intelligence (IJCAI) 2021.

[P9] **Pusalkar, N.***, Karmakar*, S., Aggrawal, R., Mali, P., Singh, A., Sarkar A., Krishna M.K., *SMA Actuated Dual Arm Flexible Gripper*, In Proceedings of the Advances In Robotics 2019 (AIR 2019), Association for Computing Machinery, New York, NY, USA, Article 59, 1-6.

[P10] Banerjee, H.*, **Pusalkar, N.***, and Ren, H., *Preliminary Design and Performance Test of Tendon-Driven Origami-Inspired Soft Peristaltic Robot*, 2018 IEEE International Conference on Robotics and Biomimetics (ROBIO), pp. 1214-1219, December 2018.

[P11] Banerjee, H., **Pusalkar, N.**, and Ren, H., *Single-Motor Controlled Tendon-Driven Peristaltic Soft Origami Robot*, ASME Journal of Mechanisms and Robotics 10(6), 064501.

[P12] Dharmadhikari V., **Pusalkar, N.**, Ghare, P., *Path loss exponent estimation for wireless sensor node positioning: Practical approach*, IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), 2018.

[P13] Patent for Humanoid Robot Swayat – Granted by The Patent Office, Government of India (Patent No.: 517158)

Research Experience

Human Machine Teaming Lab | Graduate Research Assistant 2021 – Present

Oregon State University, Corvallis, OR (Advisor: Prof. Julie A. Adams)

- Developed an event-triggered priority-driven and preemptive coalition reformation framework for heterogeneous multi-capability multi-robot systems (<100 robots) to facilitate coalition reallocations for dynamic new tasks [P2].
- Formulated coalition reformation as a multi-objective optimization problem that was solved using lexicographic lookahead beam search resulting in >95% successful new task assignments with <5% coalition disruptions [P2].
- Developed LN-GRAPE-S, a scalable hedonic game-based coalition formation algorithm for up to 10,000-robot simulated heterogeneous collectives with sub-linear time and communication complexity in collective size, and theoretical guarantees on Nash stability [P3].
- Abstracted the coalition formation problem as a leader-level hedonic game, achieving a minimal 6.55 MB communication overhead for 10,000 robots with optimal solutions, demonstrating an average 825x communication reduction and 29x runtime improvement over GRAPE-S baseline [P3].
- Developed a collective coalition formation algorithm, LeaderGRAPE-S, that integrated a leader-follower hierarchy with hedonic game properties resulting in an average 11x lower communication and 7x faster runtime than an existing baseline (GRAPE-S) for 10,000-robot simulated heterogeneous collectives [P4].
- Conducted non-parametric statistical analysis with effect size measures to establish statistical significance of algorithmic performance differences [P3][P4].
- Reviewed >100 papers on dynamic multi-robot task allocation and team reformation techniques, and proposed 13 classification criteria representing real-world conditions and algorithm characteristics [P1].
- Formulated a dependency graph explaining the interrelationships of factors affecting coupling in multi-robot systems to broaden the concept of coupling beyond task and robot-oriented perspectives.
- Developed a Decision Support System and a User Interface using Unity that provided contextual information and assisted human operator's decision making in autonomous underwater grasping missions [P5][P6].
- Integrated a Bravo 7 manipulator, a Trisect RGB-D camera, and a custom grasp synthesis framework via ROSbridge, and conducted field deployment in a wave pool to demonstrate successful operator-assisted grasping [P5][P6].

Honda Research Institute | Student Associate Jan 2024 – March 2024

Supervisor: Dr. Vaishnav Tadiparthi

- Implemented a double-round auction mechanism for dynamic coalition formation for tasks with 2 different priorities and stochastic arrival times using a multi-objective utility function that consider robot's current task assignment.
- Demonstrated ≈29% increase in task assignment success rate and ≈11% improvement in resource utilization for prioritized dynamic task allocation over the baseline.

Laboratory for Progress | Graduate Researcher 2020 – 2021

University of Michigan, Ann Arbor, MI (Advisors: Dr. Zhen Zeng, Prof. Chad Jenkins)

- Implemented baseline POMCP using Monte Carlo Tree Search (MCTS) for the multi-object search while maintaining beliefs over object locations via Bayesian updates, and graphically visualized belief propagation and belief state entropy throughout the simulation.
- Developed a dynamic graph visualizer to visualize the supporting relations among cluttered objects as received from a ROS message, and integrated it via ROSbridge with the pose estimation module.

UM Ford Center for Autonomous Vehicles | Graduate Research Assistant 2020 – 2021

University of Michigan, Ann Arbor, MI (Advisors: Dr. Xiaoxiao Du, Prof. Ram Vasudevan)

- Developed angle-regulated transformer models that accounted for shape and smoothness of trajectories to improve pedestrian trajectory prediction [P8].
- Implemented SOTA pedestrian trajectory prediction methods like SGAN, Social LSTM, Next Prediction, DAGnet, Social-STGCNN, and created a data loader for evaluating these methods on the Stanford Drone Dataset (SDD).
- Implemented crosswalk detection using template matching, Haar cascade classifier, and Laplacian of Gaussian.
- Performed Semantic Segmentation using FCNN, DeepLabv3, encoder-decoder model, SegNet.

Robotics Research Center | Research Assistant 2018 – 2019

International Institute of Information Technology (IIIT-H), Hyderabad, India (Advisor: Prof. K. Madhava Krishna)

- Developed a Shape Memory Alloy actuated flexible gripper for gripping objects with variable curvatures (25-190 mm) and different textures (wood, plastic, metal).
- Designed gripping mechanism with 1-way Shape Memory Alloy and belt interlocking mechanism with magnets.

Visvesvaraya National Institute of Technology, Nagpur | B.Tech Thesis 2018

Advisor: Prof. Pradnya Ghare

- Developed a learning-based wireless sensor node localization framework to estimate firefighters' locations using Received Signal Strength Indicator values from IRIS motes.
- Localized randomly deployed wireless sensor nodes by calculating distances between blind and anchor nodes using log distance path loss model and estimating position coordinates using a Multi-Layer Perceptron (MLP) network.

IvLabs, Visvesvaraya National Institute of Technology, Nagpur, India | Core Member 2016

Advisor: Prof. Shital Chiddarwar

20 DOF Kid-sized Humanoid Robot

- Generated walking gaits for a 20 DOF humanoid robot using the Linear Inverted Pendulum Model. Developed a ROS-based decision-making algorithm for Sprint Event at the Hurocup Competition organized by the Federation of International Robot Sport Association (FIRA) in Beijing in December 2016, being the only team to represent India.
- Assembled 20 DOF humanoid robot using Herkulex DRS 101 and DRS 201 motors connected in daisy chain, 3D printed brackets and links, and FitPC-2i as the main processor.
- Generated offline walking gaits using inverse kinematics in MATLAB and using Linear Inverted Pendulum Model while ensuring stability of Center of Mass (COM) and Zero Moment Point (ZMP).
- Proposed differential RGB colour space for robust marker detection under varying light intensity.
- Devised algorithm for pose estimation of the humanoid with respect to a custom-made two-coloured square-in-square marker using OpenCV.
- Developed decision making node to maintain straight line motion using estimated pose from marker and pose obtained from the IMU reading in ROS Indigo.
- Built 3 DOF robotic writing and grasping arms, and generated trajectories using inverse kinematics for writing alphabets and manipulating objects considering the arm's dextrous workspace.

Dual Robotic Writing and Grasping Arms Dec 2015

- Built a 3 DOF robotic writing arm capable of writing six English alphabets and 3 DOF robotic grasping arm which could grasp and move objects using Dynamixel AX-12A motors and controlled by OpenCM-9.04.
- Generated trajectories using inverse kinematics for writing alphabets and manipulating objects while considering the arm's dextrous workspace.

Nao Robot | Intern May 2016

Gade Autonomous Systems Private Limited, Mumbai, India

- Implemented image processing algorithms for marker detection and line tracking using OpenCV and Ubuntu 14.04.
- Developed independent software modules for these tasks and integrated them in the NAOqi SDK for the Nao robot.

Autonomous Robots and Multi-Robot Systems Lab | Intern Dec 2017

Indian Institute of Technology, Mumbai, India (Advisor: Prof. Arpita Sinha)

- Created a differential drive robot model having Hokuyo laser range sensor in Gazebo, set up navigation stack, and performed mapping and localization using gmapping package and AMCL node.

Medical Mechatronics Lab | Intern May 2017 – July 2017

National University of Singapore, Singapore (Advisor: Prof. Hongliang Ren)

- Fabricated a prototype of origami-inspired soft robot using Yoshimura Origami Fold pattern and demonstrated peristaltic movement using a compression spring mechanism.
- Designed corrugated origami-inspired support structures to mitigate buckling-induced non-axial deformation.

Robotics System Laboratory 2020

University of Michigan

- Designed a cascaded PID controller with successive loop closure using wheel odometry and gyrodometry for 2-wheeled Segway robot.

- Implemented occupancy grid mapping, particle filter-based Monte Carlo Localization, and A* path planning for a two wheeled differential drive robot in simulation.
- Developed trajectories for pick and place task using inverse kinematics for a 4DOF manipulator.

Vehicle Classification in Images 2019

University of Michigan

- Implemented Transfer Learning using Mask-RCNN for classification of vehicles into 4 classes achieving about 95% accuracy.

Smooth Motion Planning for Serial Manipulators with Velocity Constraints 2021

University of Michigan

- Extended RRT* and BiRRT for velocity-constrained serial manipulators by formulating a velocity-biased weighted distance metric that biases the planner to minimize trajectory execution time under joint velocity impediments.
- Proposed a minimum-time cubic spline path smoothing algorithm achieving up to 13.66% reduction in trajectory execution time over standard shortcut smoothing for BiRRT.

Turtlebot Path Planning 2023

Oregon State University

- Implemented obstacle-aware autonomous navigation and path planning for Turtlebot2 using perception from LiDAR and Kinect to estimate obstacle locations.

Kinematic Path Planning under Uncertainty 2023

Oregon State University

- Implemented A* path planning with a Jacobian-based covariant Riemannian metric derived from joint-space covariance, enabling uncertainty-aware motion planning that minimizes end-effector uncertainty along the path.
- Demonstrated that Covariant A* generates smoother paths with fewer orientation changes compared to Euclidean A* across two over-actuated planar manipulator configurations in SE(2).

Taxi Fleet Management using a Dense Reward Structure 2023

Oregon State University

- Developed a decentralized MARL framework for taxi fleet management in a stochastic urban environment, using a dense reward function with coverage, path length, and congestion penalties for learning optimal coverage policies.
- Demonstrated convergence of Q-learning policies to distinct coverage points across a fleet of up to 10 taxis in a 5x10 grid, with taxis learning congestion-minimizing paths without explicit coordination.

Multi-robot control with Natural Language 2026

Oregon State University

- Created a custom VLA by finetuning HuggingFace's SmolVLM-256M-Instruct model using LoRA (rank 16) to predict Boids parameter weights and average velocity for 8 multiple robots moving through an office environment.

Other Robotics Projects 2016

IvLabs

- Developed an algorithm to track and plot the movement of a colored marker in OpenCV.

- Built a Bluetooth controlled robot using Bluetooth module HC-05 and Android Bluetooth Application.
- Built an RF controlled bot using a 434MHz RF Module and encoder-decoder ICs for data transmission.

Professional Service

- **Reviewer:**
  - AAMAS ALA Workshop 2026
  - IEEE RA-L 2026
  - AAAI 2026 (Student Abstract and Poster Program)
  - ICAPS 2025
  - IROS 2025
  - CASE 2023
- **Leadership:** President, Artificial Intelligence Graduate Student Association, OSU, 2024-2025
- **Mentor:**
  - AI Application Support Program Mentor: Reviewed resumes and SOPs of students applying to graduate programs at OSU, 2024, 2025.
  - Participated in a student panel with 5 other mentors to provide graduate school-related advice to approximately 25 undergraduate REU students at OSU.
  - Project Mentor: Mentored a high school student to develop an image-detection framework to identify humans in natural disasters, 2022.
  - Graduate and Undergraduate Student Mentor at University of Michigan.
- **Volunteer:**
  - ICRA 2017 Student Volunteer, Singapore: Registration desk, Session organization, Lab tour guide.
  - Ivlabs: Conducted robotics and basic electronics workshops for freshmen and sophomore students.

Technical Skills

- Programming Languages: C/C++, Python, Basic Javascript and C#, MATLAB, Embedded C.
- Software and Libraries: Pytorch, ROS, RViz, OpenMPI, High Performance Computing (HPC) Cluster, OpenCV, SolidWorks, EagleCAD.
- Simulation Platforms: Gazebo, VMAS.
- Hardware: NAO Robot, Turtlebot, Embedded Systems (FitPC, Arduino, OpenCM, Atemga microcontrollers).
- Other: SLURM, Docker, Git, Linux.
- Task Allocation, Multi-Robot coordination, Heterogeneous Multi-Robot systems, Distributed Execution, Model-based Planning, Multi-Objective Optimization, Non-Parametric Statistical Analysis.

Awards and Honors

- Selected as a Future Leader of AI at the ACM AI Leadership Summit 2026.
- Awarded the AAMAS 2026 Student Scholarship.
- Received the Best Paper Award in Robotics/UAV vertical for the paper titled *Leader-based Coalition Formation for Extremely Large Scale Collectives* at IEEE Conference on Artificial Intelligence (CAI), 2025.
- Eligible for the Gov. of India's INSPIRE Scholarship for Higher Education offered to top 1 percentile students in Class 12 Board (HSC) Examination, 2014.